

On Ramanujan Challenge Problem 3.2: Optimality of Apéry's Irrationality-Measure Bound for $\zeta(3)$

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Abstract

We study the conjecture $\gcd(d_n a_n, d_n b_n) = e^{o(n)}$, where $(a_n), (b_n)$ are the Apéry sequences for the irrationality of $\zeta(3)$ and $d_n = \text{lcm}(1, \dots, n)^3$. Computational evidence for $n \leq 100$ shows that $\log(\gcd)/n$ decreases from a peak of 0.93 at $n = 6$ to 0.06 at $n = 100$, consistent with $e^{o(n)}$ growth. We analyze the p -adic structure for small primes and connect the conjecture to supercongruences of Apéry numbers.

1 The Apéry sequences

The sequences satisfy

$$(n+1)^3 u_{n+1} - (34n^3 + 51n^2 + 27n + 5) u_n + n^3 u_{n-1} = 0, \quad n \geq 1, \quad (1)$$

with $a_0 = 0, a_1 = 6, b_0 = 1, b_1 = 5$.

Setting $d_n = \text{lcm}(1, \dots, n)^3$, Apéry proved that $d_n a_n$ and $d_n b_n$ are integers. The linear form $a_n - \zeta(3) b_n \rightarrow 0$ at geometric rate gives the irrationality of $\zeta(3)$ and an irrationality measure $\mu(\zeta(3)) \leq 1 + \log(34 + 15\sqrt{5})/\log(34 - 15\sqrt{5}) \approx 5.513\dots$

2 The conjecture and its implication

Conjecture 1 (Weinbaum et al. 2025). $\gcd(d_n a_n, d_n b_n) = e^{o(n)}$.

If this holds, then Apéry's irrationality-measure bound is *tight* for his particular construction: no unexpected cancellation between numerator and denominator can improve the bound without fundamentally different rational approximants.

3 Computational evidence

We computed $\gcd(d_n a_n, d_n b_n)$ exactly for $n = 1, \dots, 100$ using exact rational arithmetic.

n	$\log(\gcd)/n$	$\gcd$
1	0.00	1
6	0.93	270
10	0.18	6
20	0.42	4590
30	0.29	6630
50	0.04	6
70	0.11	1674
100	0.06	570

The ratio $\log(\gcd)/n$ shows no systematic upward trend and decreases on average, consistent with $e^{o(n)}$.

4 p -adic analysis

For each prime $p \leq 31$, we computed the p -adic valuation $v_p(\gcd(d_n a_n, d_n b_n))$ for $n = 1, \dots, 50$. No prime contributes a linearly growing valuation:

- v_2 : bounded, values $\{0, 1\}$
- v_3 : oscillates between 0 and 3, no linear trend
- v_5 : sparse nonzero values, bounded by 1
- All primes $p \geq 7$: very sparse nonzero contributions

This is consistent with the supercongruence $a_{mp} \equiv a_m \pmod{p^3}$ (Beukers), which controls the p -adic growth of a_n and hence of the $\gcd$.

5 Connection to supercongruences

The Beukers supercongruence

$$a_{mp^r} \equiv a_{mp^{r-1}} \pmod{p^{3r}}$$

implies that the p -adic valuation of a_n (and similarly b_n) grows at most as $3\lfloor \log_p n \rfloor = O(\log n)$ in the index n . Since $d_n = e^{n+o(n)}$ by the prime number theorem, and the $\gcd$ cannot exceed $\min(|d_n a_n|, |d_n b_n|)$, the supercongruences provide the mechanism preventing exponential $\gcd$ growth.

A full proof requires two ingredients:

1. **Tower rigidity** (from supercongruences): For each prime p , the p -adic valuation along the p -power tower $n = mp^r$ is controlled by $v_p(A_{mp^r} - A_{mp^{r-1}}) \geq 3r$. This bounds the p -adic contribution per prime.
2. **Prime-counting estimate** (the missing ingredient): Show that only $o(n/\log n)$ primes can divide both $d_n a_n$ and $d_n b_n$ at index n . Equivalently, for all but a sparse set of primes p , at least one of $d_n a_n$ or $d_n b_n$ is a p -adic unit. This is the step that prevents exponential conspiracy.

The prime-counting estimate may follow from the connection between Apéry numbers and the Fourier coefficients of a weight-4 modular form (Stienstra–Beukers), via the associated ℓ -adic Galois representation and its Chebotarev-type equidistribution.

References

- [1] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$* , Astérisque **61** (1979), 11–13.
- [2] F. Beukers, *Some congruences for the Apéry numbers*, J. Number Theory **21** (1985), 141–155.